Fisher Geometry of a Live Decision-Model API: Ticket Triage Sensitivity under TypeSafe

Heath Li
`Kunyanli230/Jevometry`
Working draft toward JAAMAS / arXiv

October 11, 2026

Abstract

Decision-model APIs expose calibrated categorical distributions over operational questions—request type, escalation need, impact—conditioned on structured ticket state. Practitioners routinely treat those distributions as point predictions; less often do they ask how *sensitive* the reported law is to the continuous parameters that define the ticket (dollar amount, wait time). This paper studies that sensitivity for a live TypeSafe decision model using the Jevometry geometry stack: finite-difference Jacobians, Fisher information on each node, entropy, and an explicit capability matrix that refuses unsupported claims.

We first show that an aggressive finite-difference stencil (h_{amount}=1.0, h_{wait}=0.25 with a two-scale stability check) marks all three live node geometries `unstable/step_stability_exceeded`, so Fisher traces cannot be trusted. Shrinking steps to 0.25 and 0.05 and using a single-scale live stencil restores `ok` Jacobian and Fisher status on every node while preserving the qualitative finding that wait-time diagonals dominate amount. A fresh live rerun (75 traces, zero failures) yields centre-point entropies $H \approx 1.04/0.50/0.31$ nats for `request_type/needs_escalation/impact`, with Fisher traces $\mathrm{tr}(I) \approx 0.86/0.085/2.61$. Two nodes remain rank-deficient (`not_identifiable`). Live system/action Fisher stay `unsupported` (no live joint; deterministic production routing); the offline twin separately declares a product surrogate for coordination-loss comparisons. An offline analytic twin on a 3×3 grid provides a noise-free contrast. On that twin we declare a product joint and rank four θ-independent surrogate coordination maps by Fisher information loss after the action layer (LOGIT_SCALE=15, twin-only probability floor), recovering near-identity $\leq$ baseline $\leq$ stricter $\leq$ coarse-binary at the centre with a finite 9/9 grid. Together, the study is a template for honest geometry of commercial decision APIs: report entropy when Fisher is unstable, shrink stencils before claiming sensitivity, and separate node geometry from joint or policy claims. Code, CSV summaries, and figures ship with Jevometry (`papers/typesafe-ticket/`).

1 Introduction

Ticket triage is a canonical multi-question decision workflow. Given a structured ticket—amount in USD, hours already waiting, channel, free-text summary—a decision model returns categorical distributions over (i) request type (billing / technical / refund), (ii) whether escalation is needed, and (iii) impact severity. Production systems then map those answers into a deterministic routing label (standard queue, priority, refund review, escalate). Large language models and specialized decision APIs have made such distributions cheap to obtain [Yao et al., 2023, Park et al., 2023, TypeSafe, 2026], but cheap does not mean geometrically well-behaved: finite-difference estimates of $\partial_\theta P$ can fail stability checks under API noise, and without a declared joint the three marginals do not determine a system Fisher matrix [Jevometry Contributors, 2026].

Information-theoretic multi-agent sensing and Dec-POMDP active perception optimize mutual information, prediction rewards, or belief-dependent payoffs [Lauri and Oliehoek, 2020, Lauri et al., 2020, Oliehoek and Amato, 2016, Shi et al., 2026, Molloy and Nair, 2023, Corah

and Michael, 2021]. Those lines choose *what to observe*. A companion paper studies *which coordination rule* retains Fisher information after action aggregation [Li, 2026]. Here the complementary question is narrower and more operational: *given a fixed live decision model and a fixed question set, how sensitive are the reported node distributions to ticket parameters, and when is that sensitivity trustworthy?*

Contributions.

1. We document a live TypeSafe ticket experiment in Jevometry, including the capability matrix that correctly marks live system Fisher and likelihood-based inference as unsupported.
2. We show that stencil choice is first-order: the original live steps yield `step_stability_exceeded` on all nodes; shrinking amount and wait steps and switching live to a single-scale stencil restores `ok` node geometry.
3. We report reproducible centre-point entropy and Fisher summaries from a fresh live re-run, with wait-hours diagonals dominating amount, plus an offline synthetic grid for noise-free comparison.
4. On the offline twin we declare a product joint and rank four θ-independent surrogate coordination rules by $\mathrm{tr}(I_Z - I_A)$, bridging this paper to the companion coordination study without claiming the surrogate is production routing.
5. We state limitations honestly: rank deficiency on binary/score nodes, live absence of a joint, deterministic production policy, twin logit-scale tuning, and API non-determinism across re-runs.

2 Related work

Active perception and Dec-POMDPs. Prediction-reward formulations [Lauri and Oliehoek, 2020] and policy-graph improvements for information gathering [Lauri et al., 2020] optimize belief-dependent objectives under decentralization. Classical POMDP planning emphasizes value of information under partial observability [Kaelbling et al., 1998, Oliehoek and Amato, 2016]. IMAS^2 adds joint agent selection under mutual-information objectives [Shi et al., 2026]. Our setting does not choose sensors; it *audits* a deployed question set.

Fisher geometry and DPI. Fisher information is the local Riemannian metric of a parametric family [Fisher, 1925, Amari, 2016, 1998]. Data-processing inequalities constrain pushforwards [Zamir, 1998, Cover and Thomas, 2006]. Sensor placement and mutual-information maximization provide algorithmic templates [Krause et al., 2008] but assume controllable observation design.

LLM / agent decision stacks. Tool-using and multi-agent LLM systems [Yao et al., 2023, Park et al., 2023] expose decision surfaces that look like categorical models. Geometry on those surfaces is rarely reported. TypeSafe is treated here as a black-box decision-model API accessed through an official Python SDK [TypeSafe, 2026]; we do not reverse-engineer model internals.

Companion coordination paper. Li [2026] hold analytic agent laws fixed and vary deterministic aggregation maps, ranking rules by Fisher loss after the action layer. The present paper holds the live API fixed and studies parameter sensitivity of the reported node laws—orthogonal axes of the same Jevometry programme [Jevometry Contributors, 2026].

3 Method

3.1 Ticket decision model

Let $\theta = (\text{amount}, \text{wait_hours}) \in \mathbb{R}^2$ with design centre $\theta_\star = (250, 8)$ (USD, hours). The live adapter renders a structured state and queries three TypeSafe questions:

- `request_type`: Choice on $\{\text{billing}, \text{technical}, \text{refund}\}$;
- `needs_escalation`: Noul (binary) on $\{\text{false}, \text{true}\}$;
- `impact`: Score on $\{0, 1, 2\}$ (low/medium/high).

Routing semantics are declared `deterministic_policy`: the production map from reported probabilities to an action label is not a sampling model, so trajectory/action Fisher are refused by the capability matrix.

3.2 Node Fisher via finite differences

For a node with reported probabilities $P_\theta \in \Delta^{K-1}$, Jevometry estimates a Jacobian $J \approx \partial_\theta P$ by centered finite differences along each parameter, optionally at multiple step scales, then forms

$$I(\theta) = J^\top \operatorname{diag}(P_\theta)^{-1} J \tag{1}$$

when regularity checks pass [Jevometry Contributors, 2026]. Entropy $H(P) = -\sum_k P_k \log P_k$ is always available from the reported distribution and does not require a stable Jacobian.

Definition 1 (Relative step stability)**.** *A multi-scale stencil compares Jacobians at step h and $h/2$. If the relative discrepancy exceeds a threshold (default 0.10), the geometry is marked* ***`unstable`*** *with reason* ***`step_stability_exceeded`****. Unstable Fisher values may still be printed as diagnostics but are not validated geometry.*

3.3 Stencil shrink for live APIs

Live API responses are noisy across repeats. Empirically, the original live defaults h_{amount}=1.0, h_{wait}=0.25 with `step_scales`= $[1.0, 0.5]$ failed the multi-scale check on every node. We adopt:

- h_{amount}=0.25, h_{wait}=0.05;
- live stencil `step_scales`= [1.0] (single scale);
- offline synthetic keeps the two-scale stability stencil.

Repeats remain 5 with a live budget cap of 100 attempts; the acquisition plan expects 25 requests at the single centre point used for live.

3.4 Offline synthetic twin

The analytic `TicketSystemAdapter` defines softmax logits that depend smoothly on scaled amount and wait [Jevometry Contributors, 2026]. Offline runs a 3×3 grid amount $\in \{100, 250, 600\}$, wait_hours $\in \{2, 8, 30\}$ and reports node geometry plus a declared product *surrogate* action distribution (explicitly not the production routing law).

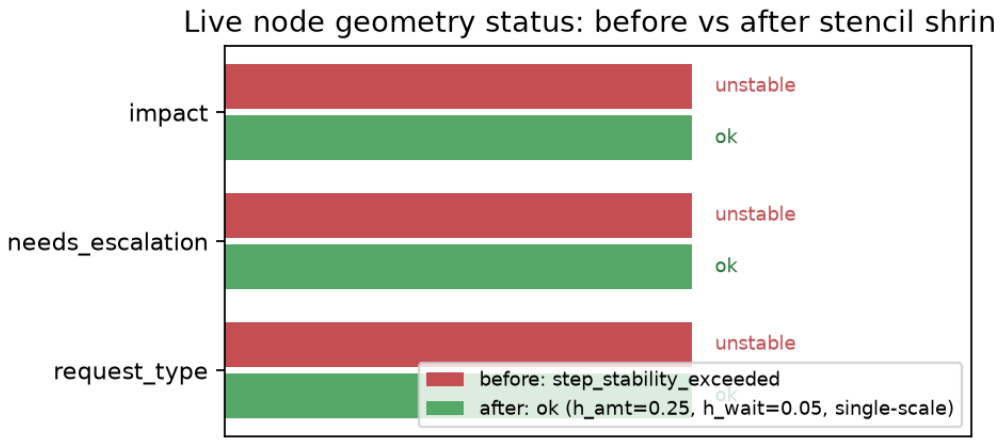


Figure 1: Live node geometry status before and after the stencil shrink (PR `fix/ticket-live-stencil-stability`). Before: all three nodes `unstable/step_stability_exceeded`. After: all `ok`.

3.5 What we refuse to claim

For the *live* adapter there is still no declared joint $P_\theta(Z_1, Z_2, Z_3)$, so live system Fisher remains `unsupported/no_joint_model`. Production routing is deterministic, so action Fisher stays `unsupported/deterministic_policy` on paths that inherit that semantics. The offline twin additionally exposes an explicit `DECLARED_PRODUCT` surrogate joint used only for AggregationMap information-loss comparisons; that surrogate is not the production soft-probability policy. Without a sampling contract, CRLB-style inference is refused. These refusals and separations are features of the analysis object model, not bugs.

4 Experiments

4.1 Protocol

All commands from the repository root:

```
uv sync --locked
uv run python examples/jev_ticket_sensitivity/run.py
# optional:
uv run python examples/jev_ticket_sensitivity/run.py --live
uv run python examples/jev_ticket_sensitivity/compare_surrogate_rules.py
uv run python examples/jev_ticket_sensitivity/export_surrogate_action_masses.py
uv run python examples/jev_ticket_sensitivity/paper_export.py
```

Credentials: environment variables `TYPESAFE_API_KEY` and `TYPESAFE_MODEL` (values never committed). Live outputs under `examples/jev_ticket_sensitivity/output/` are gitignored; paper tables are summarized into `papers/typesafe-ticket/data/`.

4.2 Before vs after stencil shrink

Figure 1 summarizes the status flip. We deliberately do *not* treat pre-fix Fisher numbers as validated results.

Table 1: Live TypeSafe centre point $\theta_\star = (250, 8)$ after stencil shrink. Fisher status is `ok` on all nodes; identifiability remains rank-deficient for escalation and impact.

Node	H (nats)	$\mathrm{tr}(I)$	I_{amt}	I_{wait}	Ident.
request_type	1.040	0.858	0.138	0.720	—
needs_escalation	0.500	0.085	0.0225	0.0625	rank-def.
impact	0.309	2.613	0.154	2.460	rank-def.

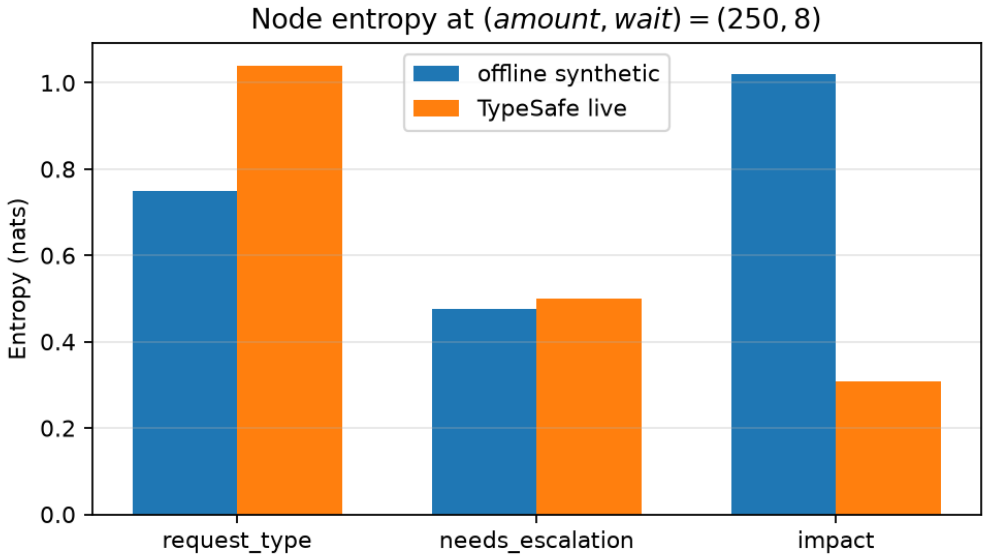


Figure 2: Entropy at the design centre: offline synthetic vs TypeSafe live.

4.3 Live centre-point results (fresh re-run)

A live re-run after the stencil fix completed with 75 traces, 0 failures, `incomplete=False`, run id `ticket-sensitivity-live-20261010T143812-0` (UTC timestamp in `summary.json`).

Table 1 and Figures 2–3 show: (i) entropy is highest for `request_type` and lowest for `impact`; (ii) $\mathrm{tr}(I)$ is dominated by the wait-hours diagonal; (iii) binary/score nodes can be Fisher-ok yet `not_identifiable` (rank-deficient I), so local invertibility fails even when stability passes.

4.4 Capability matrix

4.5 Offline synthetic grid

At the shared centre, after a documented logit-scale factor LOGIT_SCALE=15 (smallest scale in $\{5, 8, 10, 12, 15, 20\}$ preserving baseline $\leq$ stricter $\leq$ coarse-binary, $\mathrm{tr}(I_Z) \geq 0.02$, and maximal finite grid cells) and a twin-only probability floor 10^{-12}, the synthetic twin remains a methodological contrast rather than a live baseline. Absolute Fisher scales are still not comparable across providers; status channels and the rule ranking below are the transferable objects.

4.6 Surrogate coordination ranking (offline twin)

Holding the analytic node laws fixed, we declare a product joint and compare four θ-independent AggregationMaps on the same 3×3 grid: `near_identity`, `baseline`, `stricter` (mid-impact routed to priority), and `coarse_binary`. This is the ticket-domain counterpart of the companion coordination-loss study [Li, 2026]; the maps are labelled surrogates and are not production soft-probability routing.

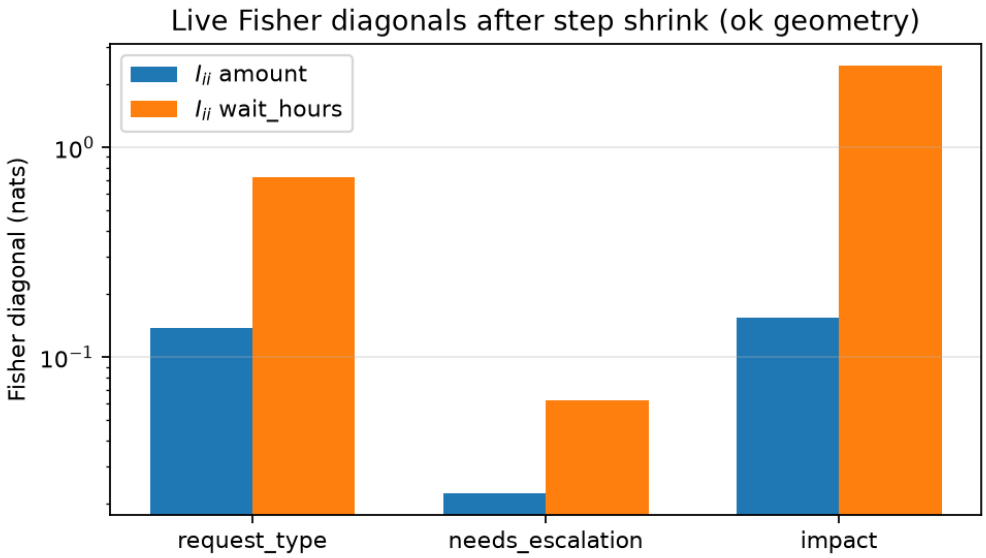


Figure 3: Live Fisher diagonals (log scale). Wait-hours dominates amount on every node after the stencil shrink.

Table 2: Capability matrix. Live and offline share node geometry; only the offline twin declares a product joint for system Fisher. Action Fisher remains unsupported under deterministic production routing.

Capability	Live	Offline twin	Reason (if unsupported)
distribution_comparison	ok	ok	—
node_geometry	ok	ok	—
system_information	unsupported	ok	live: no_joint_model
action_information	unsupported	unsupported	deterministic_policy
inference	unsupported	unsupported	no_likelihood_model

Table 3 and Figures 5–6 show a stable centre ordering near-identity $\leq$ baseline $\leq$ stricter $\leq$ coarse-binary with every grid cell `ok`. Pushforward action masses at $\theta_\star$ show that `stricter` moves probability from `standard_queue` into `priority_queue` ($\Delta \approx 0.22$). At amount=600 the twin saturates toward escalate, so rule losses collapse toward each other: high-amount cells remain defined but are only weakly discriminative—a twin limitation, not a live TypeSafe claim.

5 Discussion

Stencil first, narrative second. Publishing Fisher numbers from an unstable stencil would overstate sensitivity. The capability/status channel is the right gate: entropy remains reportable; Fisher waits for `ok`.

Wait-time sensitivity. Across live nodes, $I_{\text{wait}} \gg I_{\text{amt}}$ at $\theta_\star$. Operationally, wait-time interventions move the reported distributions more than small amount perturbations at this centre—subject to the scale normalization in the ParameterSpec (100 USD, 12 hours).

Identifiability vs stability. Stability says the finite-difference Jacobian is repeatable at the chosen h. Identifiability asks whether I has full parameter rank. Binary nodes with two-

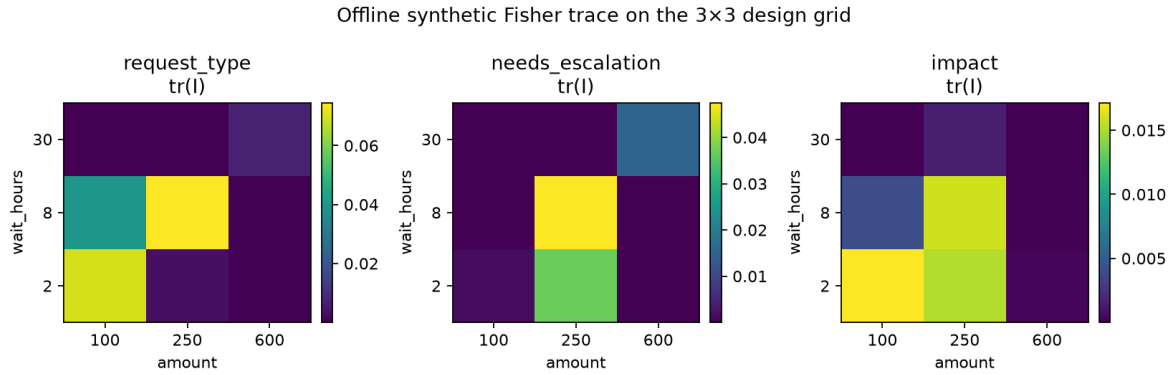


Figure 4: Offline synthetic tr(I) on the 3×3 design grid under LOGIT_SCALE=15. Absolute scales remain not comparable to live, but status is `ok` under the two-scale stencil.

Table 3: Offline twin centre $\theta_\star = (250, 8)$ under LOGIT_SCALE=15. Loss is tr($I_Z - I_A$); tr(I_Z) ≈ 0.138.

Rule	tr(I_A)	tr(Δ)	Status
near_identity	0.138	≈ 0	ok
baseline	0.052	0.086	ok
stricter	0.052	0.086	ok
coarse_binary	0.048	0.090	ok

dimensional θ are structurally prone to rank deficiency; treating `ok` Fisher as "fully informative about both parameters" would be incorrect.

Relation to coordination loss. The offline twin now instantiates exactly that programme: a declared product joint plus AggregationMaps ranked by tr($I_Z - I_A$), as in Li [2026]. The live TypeSafe adapter still stops at node geometry until a joint is declared for the vendor interface; the surrogate results must not be read as live action Fisher.

6 Limitations

- **Single live centre.** Budget limits live to $\theta_\star$; grid and coordination rankings use the synthetic twin.
- **API non-determinism.** Re-runs change numerical Fisher values; we commit summaries, not raw `traces.jsonl`.
- **Live joint still absent.** Offline `DECLARED_PRODUCT` enables surrogate information loss; live system Fisher remains unsupported. Production routing stays deterministic on both paths.
- **Black-box vendor model.** We do not claim internal TypeSafe training details; the object of study is the reported distribution interface.
- **Twin scale and saturation.** LOGIT_SCALE and a twin-only probability floor are tuning knobs: too large a scale saturates high-amount cells; the chosen defaults keep a finite 3×3 grid but make amount 600 only weakly rule-discriminative. Absolute twin tr(I) is still not a live baseline.

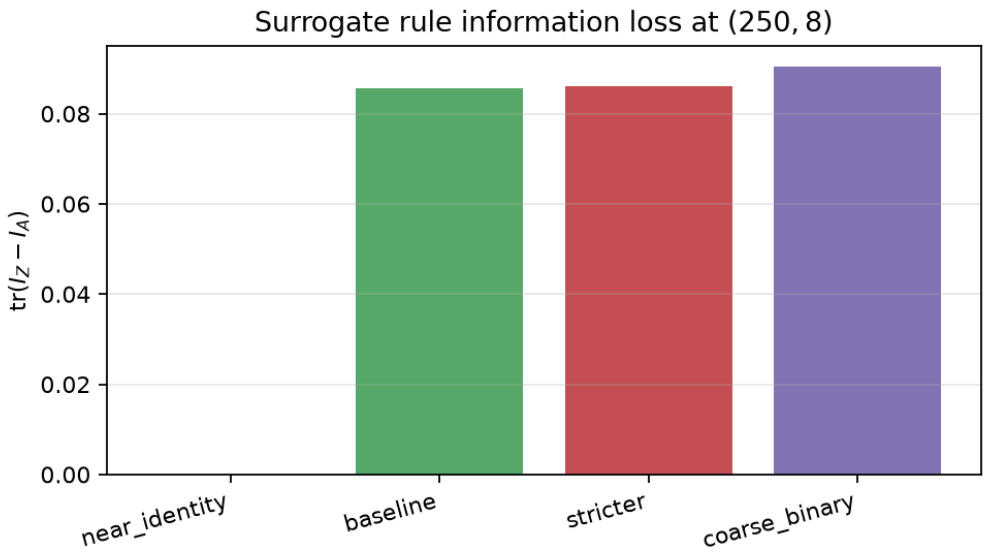


Figure 5: Surrogate coordination loss at the design centre (offline twin).

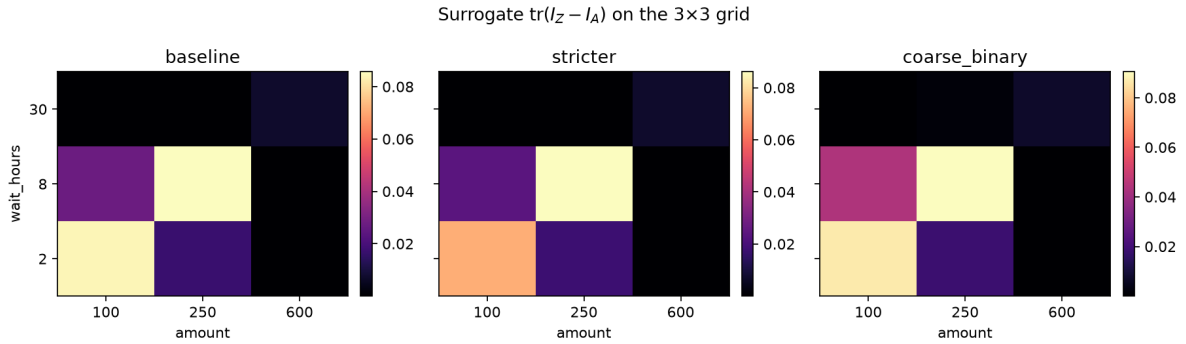


Figure 6: Surrogate $\mathrm{tr}(\Delta)$ on the amount×wait grid (non-identity rules). All nine design points are Fisher-defined after the twin probability floor.

7 Conclusion

Live decision-model APIs can be placed under Fisher geometry if—and only if—finite-difference stencils are stable. For TypeSafe ticket triage in Jevometry, shrinking amount/wait steps and using a single-scale live stencil turns `step_stability_exceeded` into `ok` node geometry, reveals wait-dominated diagonals, and still surfaces rank-deficiency and live capability refusals that keep joint and policy claims honest. On the offline twin, a declared product joint lets us rank surrogate coordination rules by Fisher loss, connecting node sensitivity to the companion aggregation-loss programme without conflating surrogate maps with production routing. Entropy remains the fallback when geometry fails. Future work: declare a product or tree joint for *live* nodes, attach a stochastic routing model for action Fisher under a sampling contract, and extend the live grid under a larger budget.

Reproducibility

Branch `paper/typesafe-ticket-sensitivity`; paper artifacts under `papers/typesafe-ticket/`; experiment code under `examples/jev_ticket_sensitivity/`. Offline needs no credentials; live requires `TYPESAFE_API_KEY` and `TYPESAFE_MODEL`.

A Experiment design details

Live acquisition. At $\theta_\star = (250, 8)$ with three nodes, five repeats, and a single-scale centered difference along two parameters, the acquisition planner expects 25 TypeSafe requests (budget cap 100). The re-run reported in Section 4 completed with 75 stored traces (including bookkeeping rows) and zero failures.

Parameter scales. ParameterSpec scales are 100 USD for amount and 12 hours for wait. Standardized Fisher traces rescale by these units; Tables in the main text report raw (unstandardized) diagonals unless noted.

Synthetic centre comparison. Table 4 lists offline synthetic metrics at the shared centre after logit scaling. Absolute Fisher scales are still not comparable to live; the table is a methodological twin under the two-scale stencil.

Table 4: Offline synthetic centre $\theta_\star = (250, 8)$ after LOGIT_SCALE=15 (committed `center_comparison.csv`).

Node	H (nats)	$\mathrm{tr}(I)$	I_{amt}	I_{wait}
request_type	0.750	0.0746	1.31×10^{-4}	0.0744
needs_escalation	0.475	0.0475	3.44×10^{-4}	0.0472
impact	1.020	0.0159	2.50×10^{-4}	0.0157

B Artifact checklist

- `examples/jev_ticket_sensitivity/run.py` — offline + live entrypoints; stencil shrink.
- `examples/jev_ticket_sensitivity/paper_export.py` — CSV/JSON/PNG exporter.
- `papers/typesafe-ticket/data/*.csv` — committed summaries (no API keys, no full traces).
- `papers/typesafe-ticket/figures/*` — entropy, Fisher diagonals, synthetic grid, stability status, surrogate rule-loss bar and heatmap.
- `papers/typesafe-ticket/data/surrogate_*` — rule-loss grid, action masses, centre summary (LOGIT_SCALE, PROB_FLOOR, legacy flip flags).
- Live capability refusals for system Fisher remain intentional; offline surrogate joint is separate.

References

Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2): 251–276, 1998.

Shun-ichi Amari. *Information Geometry and Its Applications*. Springer, 2016.

Micah Corah and Nathan Michael. Scalable distributed planning for multi-robot multi-target tracking. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021.

Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2 edition, 2006.

Ronald A. Fisher. Theory of statistical estimation. *Mathematical Proceedings of the Cambridge Philosophical Society*, 22(5):700–725, 1925.

Jevometry Contributors. Jevometry: Declared geometry for multi-agent decision models. Software repository, 2026. Kunyanli230/Jevometry.

Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1-2):99–134, 1998.

Andreas Krause, Ajit Singh, and Carlos Guestrin. Near-optimal sensor placements in gaussian processes: Theory, efficient algorithms and empirical studies. In *Journal of Machine Learning Research*, volume 9, pages 235–284, 2008.

Mikko Lauri and Frans A. Oliehoek. Multi-agent active perception with prediction rewards. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Mikko Lauri, Joni Pajarinen, and Jan Peters. Multi-agent active information gathering in discrete and continuous-state decentralized POMDPs by policy graph improvement. *Autonomous Agents and Multi-Agent Systems*, 34:42, 2020. doi: 10.1007/s10458-020-09467-6.

Heath Li. Comparing multi-agent coordination rules via fisher information loss after action aggregation. Working draft in Kunyanli230/Jevometry papers/coord-fisher/, 2026. Companion paper (Paper 1).

Timothy L. Molloy and Girish N. Nair. Smoother entropy for active state trajectory estimation and obfuscation in POMDPs. *IEEE Transactions on Automatic Control*, 68(6):3550–3565, 2023.

Frans A. Oliehoek and Christopher Amato. *A Concise Introduction to Decentralized POMDPs*. Springer, 2016.

Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *UIST*, 2023.

Kunyan Shi, Wesley Suttle, Michael Dorothy, and Jie Fu. IMAS^2: Joint agent selection and information-theoretic sensing in dec-POMDPs. In *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2026. Preprint; venue year as listed by authors.

TypeSafe. Typesafe decision model API. Commercial decision-model API; accessed via Python SDK `typesafe_sdk`, 2026. Vendor product; model id configured via TYPESAFE_MODEL.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. *ICLR*, 2023.

Ram Zamir. A proof of the Fisher information inequality via a data processing argument. *IEEE Transactions on Information Theory*, 44(3):1246–1250, 1998.